

Academic Lab Sizing & Baseline Assumptions Document

This document outlines the operational sizing calculations and architectural assumptions for the SalesOps AI cloud infrastructure. Designed as an academic lab deployment in the US East (N. Virginia) (us-east-1) region, these derivations mathematically justify the baseline parameters configured within the companion AWS Pricing Calculator configuration.

1. Core Target Roles & User Scale

The infrastructure sizing model is dimensioned around an active deployment serving a baseline of 1,000 Monthly Active Users (MAU). The system targets two primary operational roles: **Reps (Sales or Service Representatives)**: Users who participate in interactive customer simulation exams and receive AI-driven coaching and feedback. **Managers (Training Managers or Admins)**: Users responsible for creating personas, simulation scenarios, and reviewing analytics dashboards.

2. Architectural Sizing Calculations

A. Amazon API Gateway & AWS Lambda Invocations

The application exposes REST API endpoints through Amazon API Gateway connected to AWS Lambda serverless compute handlers. During active simulation exams, the client application performs periodic background synchronization requests to monitor session state and retrieve inbox events.

Operational Assumptions: Each representative performs an average of 5 active simulation exams per month. Each exam session lasts approximately 180 seconds (3 minutes). The frontend executes a synchronization request every 2 seconds during active sessions. **Metric Derivation:** $180 \text{ seconds} / 2\text{-second polling interval} = 90$ synchronization requests per exam. Approximately 110 additional API transactions occur per exam for authentication, scenario retrieval, answer submission, evaluation, and analytics processing. Total: 200 API transactions per exam session. $1,000 \text{ users} \times 5 \text{ exams} \times 200 \text{ API transactions} = 1,000,000$ monthly API requests and Lambda invocations. Lambda sizing is configured at 512 MB memory allocation.

B. Amazon DynamoDB Database Provisioning

The system stores operational data across several entities including users, personas, scenarios, exam sessions, evaluations, and analytics records.

Storage Assumptions: Each active user generates approximately 1 MB of JSON-based operational and evaluation data per month. $1,000 \text{ users} \times 1 \text{ MB} = 1 \text{ GB}$ raw data storage. An additional overhead multiplier is included for indexes, archived records, and semester-long retained datasets, resulting in a 5 GB allocation target. **Throughput Assumptions:** Continuous polling operations generate read activity. Answer submissions and evaluation updates generate write activity. The system therefore assumes 1,000,000 read requests and 1,000,000 write requests per month.

C. Amazon SQS Asynchronous Queue Sizing

The architecture uses Amazon SQS to support asynchronous processing and delayed inbox-style message delivery during active simulations.

Each simulation interaction generates queue operations including message creation, polling, and deletion, resulting in an estimated 1,000,000 SQS requests per month.

D. AWS Secrets Manager Token Record Allocation

To maintain secure serverless deployment practices, application secrets and external API credentials are separated entirely from source code. AWS Secrets Manager is used to securely store authentication tokens, deployment secrets, and external OpenAI API credentials powering the AI evaluation engine.

The infrastructure sizing assumes 5 securely stored secrets.

3. AWS Pricing Calculator Mapping Summary

AWS Infrastructure Component	Sizing Metric Cap	Provisioned Configuration
Amazon API Gateway	HTTP REST API Requests	1,000,000 Requests / Month
AWS Lambda	Serverless Compute Invocations	1,000,000 Invocations / Month (512 MB)
Amazon DynamoDB	Payload Throughput Units	1,000,000 Read / 1,000,000 Write Requests
Amazon DynamoDB	Cumulative Data Storage	5 GB Allocated Storage Volume
Amazon SQS	Asynchronous Queue Messages	1,000,000 Requests / Month
AWS Secrets Manager	Cryptographic Secret Records	5 Stored Secrets